

torch.nn.utils.clip_grads_with_norm_#

https://pytorch.org/docs/stable/generated/torch.nn.utils.clip_grads_with_norm_#

Description:

Scale the gradients of an iterable of parameters given a pre-calculated total norm and desired max norm. The gradients will be scaled by the following calculation Gradients are modified in-place. Note: The scale coefficient is clamped to a maximum of 1.0 to prevent gradient amplification. This ensures that gradients are only scaled down when the total norm exceeds max_norm. This function is equivalent to `torch.nn.utils.clip_grad_norm_()` with a pre-calculated total norm.

Function Signature

```
torch.nn.utils.clip_grads_with_norm_(parameters, max_norm,
total_norm, foreach=None)[source]#
```

Parameters

Returns

Return type

Returns:

None

Detailed Documentation

Documentation

Scale the gradients of an iterable of parameters given a pre-calculated total norm and desired max norm.

The gradients will be scaled by the following calculation

Gradients are modified in-place.

Note: The scale coefficient is clamped to a maximum of 1.0 to prevent gradient amplification. This ensures that gradients are only scaled down when the total norm exceeds max_norm.

This function is equivalent to `torch.nn.utils.clip_grad_norm_()` with a pre-calculated total norm.

parameters (Iterable[Tensor] or Tensor) – an iterable of Tensors or a single Tensor that will have gradients normalized

max_norm (float) – max norm of the gradients

total_norm (Tensor) – total norm of the gradients to use for clipping

foreach (bool) – use the faster foreach-based implementation. If None, use the foreach implementation for CUDA and CPU native tensors and silently fall back to the slow implementation for other device types. Default: None

